



Analysis Data Model (ADaM) Oncology Examples

Version 1.0 (Provisional)

Prepared by the CDISC ADaM Oncology Team

Notes to Readers

- This is Version 1.0 of the ADaM Oncology Examples.
- This document is based on SDTM v1.4, SDTMIG v3.2, ADaM v2.1, and ADaMIG v1.2..

See [Appendix B](#) for Representations and Warranties, Limitations of Liability, and Disclaimers.

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1 Introduction

The Analysis Data Model (ADaM) Oncology Examples was developed by the ADaM Oncology Subteam, which was directed by the ADaM Standards Team to review the use of the Model and its implementation guide (ADaMIG) in the development of clinical trials focused on oncology treatments. ADaM and the ADaMIG are available at: <https://www.cdisc.org/standards/foundational/adam>.

CDISC therapeutic area user guides (TAUGs) may also use this document when describing common methods used in oncology trials. The ADaM Oncology Examples enable TAUG development teams to focus on analytics that are specific to those therapeutic areas.

This document was created based on use of the Study Data Tabulation Model (SDTM) Version 1.4 and the SDTM Implementation Guide for Human Clinical Trials (SDTMIG) Version 3.2 as the source data for the analytics. The analytic data models were created based on ADaM Version 2.1 and ADaMIG Version 1.2.

Although this document provides examples of SDTM domains, ADaM results metadata, ADaM variable metadata, and ADaM data, these examples are provided to demonstrate how these models can be used to support regulatory submissions. This document does not contain prescriptive or binding guidance.

1.1 Purpose

Analysis of data for clinical trials used to support submissions for oncology therapies often requires some unique approaches in using ADaM. The purpose of this document is to provide descriptions and examples of how these approaches to ADaM can be applied in clinical trials. The sections of this document are broken down by the reasonable datasets used. Each section has been created by a team of individuals with experience in clinical trials used to support submissions of oncology therapies.

Each section of this document

- describes a specific approach to creating an analysis, and discusses why the analysis method is appropriate;
- provides examples of the tables to be created and the analysis datasets and metadata used; and
- provides specific references to controlled terminology available for that analysis.

On occasion, a concept or variable which is not represented in the current ADaMIG may be described in this document. Any concept that is not covered in the ADaMIG is identified as **new** in the example in which it is used, and identified as being provisional. The [Provisional Content](#) section identifies all such concepts and variables used in this document.

1.2 Organization of this Document

This document is organized in the following manner:

- Section 1, [Introduction](#), provides an overview of application of the ADaM standard to oncology clinical trials and how this document relates to other CDISC data standards.
- Section 2, [Subject-level Characteristics](#), covers the creation of the subject-level analysis dataset (ADSL) along with key baseline variables.
- Section 3, [Cycle Derivation \(ADCYCLE\)](#), describes the creation of an interim dataset used for determining the timing of each dosing cycle.
- Section 4, [Laboratory Results \(ADLB\)](#), covers topics on the creation of laboratory datasets following the ADaM Basic Data Structure (BDS).
- Section 5, [Study Drug Exposure \(ADEXP and ADEXSUM\)](#), describes the creation of analysis datasets for study drug exposure.
- [Appendices](#) provide additional background material and describe other supplemental material relevant to implementation.

1.3 Controlled Terminology

CDISC Controlled Terminology is a set of standard value lists that are used throughout the clinical research process, from data collection through analysis and submission. Terminology is updated twice a year by the CDISC Terminology Team and is published by the National Cancer Institute's Enterprise Vocabulary Services (NCI EVS) and is available at: <http://www.cancer.gov/cancertopics/>.

1.4 Provisional Content

In some cases the methods in this document may use concepts or variables that are not documented in the ADaMIG but are under consideration by the ADaM Standards Team for future inclusion. These concepts or variables may cause compliance issues until resolved by the ADaM Standards Team and their use must be documented within the reviewer's guide:

- PARQUAL. This variable is used as a method to further qualify the PARAM and PARAMCD variables. As of the time of publication of this document, the use of PARQUAL is requested by the US Food and Drug Administration (FDA) per OCE/OD documentation.

2 Subject-level Characteristics

2.1 Purpose - Subject-level Characteristics

Individual subject characteristics are important for the analysis of oncology data and there is great interest in understanding the initial diagnosis, cancer staging, prior disease progression, pathology, prior treatments or therapies, and performance status at baseline. These subject characteristics play a vital role in determining treatment options and evaluating patient response to treatments. This type of information in the analysis datasets is usually presented in the ADaM subject-level Analysis Dataset (ADSL).

At present, ADaM does not define standard variables for oncology therapeutic areas but recommends that sponsors create specific subject-level variables following the ADaM guidelines. Sponsor-defined variables may work well within a given submission, although there is a need for industry-wide standardization. The purpose of this document is to explain the concepts behind these subject-level characteristics in the context of analysis, as well as to propose standard variables for common cancer characteristics along with the associated metadata that is aligned with the ADaM guidelines and CDISC oncology TAUGs (available at <https://www.cdisc.org/standards/therapeutic-areas>). Additional subject-level variables required for interim analysis are also discussed.

2.2 Subject-level Variable and Parameter Naming Fragments

The following table lists the common fragments in oncology. It is recommended to use these fragments in naming subject-level variables, whenever the concept applies. In addition, fragments defined in the SDTM and the SDTMIG must be considered to avoid creating an ADaM variable with the same name as an SDTM dataset variable.

Fragment	Fragment Label	CDISC Notes
ACT	Anti-cancer Therapy	The position of this fragment within the variable name depends on the purpose of the variable.
DX	Diagnosis	The position of this fragment within the variable name depends on the purpose of the variable.
PD	Progressive Disease	The position of this fragment within the variable name depends on the purpose of the variable.
MS	Metastasis	The position of this fragment within the variable name depends on the purpose of the variable.
CHM	Chemotherapy	The position of this fragment within the variable name depends on the purpose of the variable.
RAD	Radiotherapy	The position of this fragment within the variable name depends on the purpose of the variable.
SRG	Surgery	The position of this fragment within the variable name depends on the purpose of the variable.
*STAGE	Stage	Suffix used when describing study-specific stage of cancers
*STT	Status	Suffix used when describing study-specific status, including substance use status
*BL	Baseline	Suffix used when variable identified is specifically a baseline value
HISTOLG*	Histopathology	Prefix used to indicate study-specific histopathology
*STYP	Subtype	Suffix used for subtypes
DISF	Disease Finding	The position of this fragment within the variable name depends on the purpose of the variable.
IMPTYP*	Immunophenotype	Prefix used to indicate immunophenotype
P	Prior	The position of this fragment within the variable name depends on the purpose of the variable.
TS*	Time since	Prefix used to indicate time since

2.3 Subject-level Variables for ADSL

This section lists important subject-level characteristic variables. As previously noted, subject-level data is usually presented in ADSL, retaining all subject-level baseline variables in ADSL as well as key covariate variables. If all baseline variables are included in ADSL, the ADSL dataset may become a very large (wide) dataset. It is possible to put baseline values in a separate ADaM BDS dataset to reduce the size of ADSL. This section describes the subject-level data in the ADSL dataset along with the BDS Analysis Dataset for Baseline Data (ADBASE) for purposes of demonstrating this approach.

2.3.1 Initial Diagnosis

For clinical trials, data collected upon initial diagnosis of cancer may include the following variables.

XX Variable Metadata

Variable Name	Variable Label	Type	Controlled Terms or Format	CDISC Notes
IDXDT	Date of Initial Diagnosis	Num		This is the date of initial diagnosis for the indication of interest. Usually this information is provided in the Medical History (MH)/SUPPMH domains.
IDXAGE	Age at Initial Diagnosis	Num		Age used for analysis that is derived from the date of initial diagnosis (IDXDT) and DM.BRTHDTC or as carried from the DM.AGE and DM.AGEU variables.
DDURN	Duration of Disease at Study Entry	Num		Actual number of units from the Initial diagnosis for the subject to the date of study entry in the clinical trial. This difference may be calculated as the duration between reference start date (e.g., randomization date, enrollment date, screening date) and date of initial diagnosis. Details for this calculation are defined in the protocol.
DDURU	Duration of Disease at Entry Units	Text		Units used for DDURN variable. This can be days, weeks, or years depending on study design.

2.3.2 Cancer Staging

The cancer stage is determined based on the primary tumor and the extent of cancer spread within the body at screening. It is common to use tumor-node-metastasis (TNM) methodology from the American Joint Committee on Cancer (AJCC; see https://training.seer.cancer.gov/collaborative/intro/systems_review.html) to assign stages to a cancer that indicate severity or extent of cancer at screening. Cancer staging is an important factor for cancer prognosis, population-based screening, and early detection. In TNM, *T* stands for extent of the primary tumor, *N* stands for regional lymph node involvement, and *M* stands for distant metastasis. The following table presents the metadata for these variables.

XX Variable Metadata

Variable Name	Variable Label	Type	Controlled Terms or Format	CDISC Notes
TSTAGE	Stage of Primary Tumor	Char	TX, T0, T1, T1A, T1B, T2, T2A, T2B, T3, T3A, T3B, T4, T4A, T4B	This is per TNM classification pertaining to primary tumor stage. Note that primary tumor stage is collected in SDTM and can have codelist that reflects the values of SDTM. It is anticipated that TNM staging will be represented in the Disease Response and Clin Classification (RS) domain. TX indicates that staging cannot be assessed. The following indicates increasing size or extent of primary tumor involvement in the order it is given: TX, T0, T1, T1A, T1B, etc.
NSTAGE	Stage of Nodal Involvement	Char	NX, N0, N1, N2, N3	Stage of nodal Involvement is collected in SDTM and can have codelist that reflects the values of SDTM. It is anticipated that TNM staging will be represented in the RS domain. NX indicates that staging cannot be assessed; N0 indicates no evidence for lymph node metastasis. N1, N2, and N3 indicate increasing extent of nodal involvement.
MSTAGE	Stage of Distant Metastasis	Char	MX, M0, M1, M1B	Stage of distant metastasis Involvement is collected in SDTM and can have codelist that reflects the values of SDTM. It is anticipated that TNM staging will be represented in the RS domain. MX indicates that staging cannot be assessed. M0 indicates no evidence for distant metastasis. M1 indicates distant metastasis and increase but below M1B. For further explanation of these terms, see the AJCC staging systems.
OSTAGE	Stage of Overall Cancer	Char	STAGE 1, STAGE 2, STAGE 3, STAGE 4, IA, IB, IIA, IIB, IIIA, IIIB, IV	This is the stage of cancer that the patient was assessed with at the point of inclusion in the study. Stage is normally captured in an SDTM domain. It is anticipated that staging will be represented in the RS domain. The codelist can be a custom codelist that reflects what is in the SDTM domain. It can have as its controlled terminology "STAGE 1," "STAGE 2," "STAGE 3," "STAGE 4," or it can use the TNM classification: IA, IB, IIA, IIB IIIA, IIIB, IV.

Variable Name	Variable Label	Type	Controlled Terms or Format	CDISC Notes
AOSTAGE	Stage of Overall Cancer Per Ann Arbor	Char	STAGE 1, STAGE 2, STAGE 3, STAGE 4, etc.	Ann Arbor staging is the staging system for lymphomas.
COSTAGE	Stage of Overall Cancer Per Cotswolds	Char	STAGE 1, STAGE 2, STAGE 3, STAGE 4, etc.	Cotswolds staging provides classification for Hodgkin lymphoma.
OSTAGE	{Stage of Overall Cancer}	Char	STAGE 1, STAGE 2, STAGE 3, STAGE 4	If a study has more than 1 staging method for overall cancer, it is suggested to use OSTAGE but prefix with a letter indicative of a method.
STAGE	{Stage of}	Char		If a study has more anatomical location or hematologic involvement, it is suggested to use STAGE but prefix with a letter indicative of anatomical location.

2.3.3 Pathology

Pathology data includes carcinoma detection, grading, and assessment of biomarkers using molecular biology methodologies.

Assessment of biomarkers using microscopy methods is usually presented with grading or positive/negative in the Microscopic Findings (MI) domain. Biomarker data variable names should follow the convention of having the biomarker indicator in the first 5 or fewer characters, followed by a suffix of "STT".

XX Variable Metadata

Variable Name	Variable Label	Type	Controlled Terms or Format	CDISC Notes
HER2STT	HER2 Status	Char	1+, 2+, 3+, Positive, Negative	Usually HER2 biomarker data by microscopy is collected in the MI domain. For example, in the initial HER2 assessment of the subject, HER2STT is equal to MI.MISTRESC when MI.MITEST represents HER2 status. MI.MITEST is subject to sponsor-controlled terminology.
ERSTT	Estrogen Receptor Status	Char	1+, 2+, 3+, Positive, Negative	This is the initial ER assessment of the subject. For example, ERSTT is equal to MI.MISTRESC when MI.MITEST represents estrogen receptor status. MI.MITEST is subject to sponsor-controlled terminology.
PRSTT	Progesterone Receptor Status	Char	1+, 2+, 3+, Positive, Negative	This is the initial PR assessment of the subject. For example, PRSTT is equal to MI.MISTRESC when MI.MITEST represents progesterone receptor status. MI.MITEST is subject to sponsor-controlled terminology.
TNBCSTT	TNBC Status	Char	Positive, Negative, Missing	This is for capturing the triple negative breast cancer status. For example, if HER2STT, PRSTT, and ERSTT are all "Negative", then TNBCSTT="Positive"; else if all 3 are missing or "UNKNOWN", then "Missing"; else "Negative".
CDX2STT	CDX2 Status	Char	Positive, Negative	CDX2 status of the subject at randomization. Lack of CDX2 expression has been shown to have poor prognostic significance for 5-year survival in certain cancers. The test is conducted on tumor samples using immunohistochemistry (IHC) methodology. For example, CDX2STT=MI.MISTRESC when MI.MITEST represents CDX2 status.
PDL1STT	PDL1 Status	Char	Positive, Negative	Initial PD-L1 status of the subject. For example, when MI.MITEST represents PD-L1 status, if MI.MISTRESN=0 then PDL1STT="Negative"; else if MI.MISTRESN>0 then PDL1STT="Positive".
*STT	{Status}	Char		Biomarker levels that are either presented with grading, positive/negative, or a missing status.

2.3.4 Molecular Markers by Laboratory

Laboratory assessments that have continuous data such as prostate-specific antigen (PSA), circulating tumor cell (CTC), or testosterone are conducted at baseline for cancer subjects.

XX Variable Metadata

Variable Name	Variable Label	Type	Controlled Terms or Format	CDISC Notes
PSABL	Baseline PSA (unit)	Num		For example, LB.LBSTRESN where LB.LBTESTCD="PSA" and LB.LBLOBXFL="Y".

Variable Name	Variable Label	Type	Controlled Terms or Format	CDISC Notes
CTCBL	Baseline Circulating Tumor Cells (Unit)	Num		For example, LB.LBSTRESN where LB.LBTESTCD="CTC" and LB.LBLOBXFL="Y".
TESTOSBL	Baseline Testosterone (Unit)	Num		For example, LB.LBSTRESN where LB.LBTESTCD="TESTOS" and LB.LBLOBXFL="Y".
*BL	Baseline {Laboratory Test} (Unit)	Num		Numeric laboratory data at baseline

2.3.5 Overall Histological Classification

The following variables define subject data indicating histological classification at diagnosis.

XX Variable Metadata [HISTOLGy]

Variable Name	Variable Label	Type	Controlled Terms or Format	CDISC Notes
HISTOLGy	Histopathology y	Char		HISTOLGy where y can be 1 to 9. Expect that 1 is the primary histologic assessment. Examples include adenocarcinoma, squamous adenosquamous, and so on.

2.3.6 Prior Disease Progression

Prior disease progression serves as an important factor to evaluate investigational drug response. Type of disease progression is indication specific. For example, in metastatic castration-resistant prostate cancer (mCRPC), disease progression is evaluated by serum PSA levels or by imaging or both.

XX Variable Metadata

Variable Name	Variable Label	Type	Controlled Terms or Format	CDISC Notes
PDTYPBL	Baseline Progressive Disease Type	Char	PSA, Radiographic/Subject to indication specific	This is the progression type that the patient was assessed with at the point of inclusion in the study. The codelist can be a custom codelist that reflects what types of progression are of interest in the study. For example, at the baseline visit progression type = PSA: If RS.RSTESTCD="TMRESP" and RSSTRESC="PD" and RS.RSTESTCD="RDIORESP" and RSSTRESC="NON-PD" and VISIT="SCREENING". Radiographic: If RS.RSTESTCD="RDIORESP" and RSSTRESC="PD" and VISIT="SCREENING".
PPDDT	Prior Progressive Disease Date	Date		This is the date that the last progressive disease was assessed prior to the current study.

2.3.7 Prior Treatment/Therapies

If the subjects are not newly diagnosed with cancer when they join the study, any prior antineoplastic therapies (i.e., anti-cancer treatments) may be of special interest to the study. Even if subjects are newly diagnosed, it may still be relevant to collect prior treatments for any other cancers they may have had.

XX Variable Metadata

Variable Name	Variable Label	Type	Controlled Terms or Format	CDISC Notes
PRADIAFL	Prior Radiotherapy Flag	Char	Y, N	Set to "Y" if there is at least 1 record with PR.PRCAT="RADIATION" and PR.PRSTDTC is prior to DM.RFXSTDTC, otherwise set to "N".
PSRGFL	Prior Surgery Flag	Char	Y, N	Set to "Y" if there is at least 1 record with PR.PRCAT="SURGERY" and PR.PRSTDTC is prior to DM.RFXSTDTC, otherwise set to "N".
PCSRGFL	Prior Cancer Surgery Flag	Char	Y, N	Set to "Y" if there is at least 1 record with PR.PRCAT="SURGERY" and some identifier that the surgery was cancer related. Otherwise set to "N".

Variable Name	Variable Label	Type	Controlled Terms or Format	CDISC Notes
PCHMFL	Prior Chemotherapy Flag	Char	Y, N	Set to "Y" if there is at least 1 record with PR.PRCAT="CHEMOTHERAPY" and PR.PRSTDTC is prior to DM.RFXSTDTC, otherwise set to "N".
PTRTEDT	Prior Treatment End Date	Num		If a patient has had prior cancer treatment, capture the date that the last treatment ended. For example: latest CM.CMENDTC where CM.CMCAT="ANTINEOPLASTICTHERAPY" or latest PR.PRENDTC where PR.PRCAT="RADIATION THERAPY" where --DTC variables are converted to numeric date format. The date part of CM.CMENDTC is captured as a numeric variable.

Note: If a study needs flags for concurrent or follow-up treatment, the prefix can be modified.

2.3.8 Performance Status at Baseline

Performance status at baseline is one of the most important factors in evaluating a subject's eligibility to enroll in investigational therapy and also in predicting how patients respond to treatment. The performance status at baseline is determined by the study team and may be different than the values collected for randomization, such as through an interactive response technology (IRT) system. Performance status is a score that estimates the patient's ability to perform certain daily activities.

There are different methods for estimating performance status (e.g., ECOG Performance Status scale, KPSS, WHO Performance Status scale).

XX Variable Metadata

Variable Name	Variable Label	Type	Controlled Terms or Format	CDISC Notes
ECOGBL	Baseline ECOG Performance Status	Num	0, 1, 2, 3, 4, 5	For example: RS.RSSTRESN where RS.RSTESTCD="ECOG101" and RSBLFL="Y".
KPSBL	Baseline Karnofsky Performance Status	Num	0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100	The KPSS allows patients to be classified as to their functional impairment. Some studies use this scale to compare the effectiveness of treatments and assess the prognosis. For example, RS.RSSTRESN where RS.RSTESTCD="KPSS01" and RSBLFL="Y".
WHOBL	Baseline WHO Performance Status	Num	0, 1, 2, 3, 4, 5	The WHO Performance Status scale is used to assess subjects' activity levels and used similarly to other scales.

Note: If a study uses different scales, then the prefix can be modified.

2.3.9 Substance Use

Substance use or abuse is an important factor and plays an important role in pathogenesis of cancer and also in predicting treatment effects.

XX Variable Metadata

Variable Name	Variable Label	Type	Controlled Terms or Format	CDISC Notes
SMOKSTT	Smoking Status	Char		Note that a subject may have multiple records in SU. If there is only 1 record with SUTRT="TOBACCO" and SUOCCUR="N", then "Never Smoker". If there is at least 1 record with SUTRT = "CIGARETTE" or "OTHER" and SUENRF="DURING/AFTER", then "Current Smoker". If all records contain SUTRT = "CIGARETTE" or "OTHER" and SUENRF="BEFORE", then "Former Smoker".
*STT	{Status}	Char		Substance abuse status examples include alcohol status and drug use status. The variable must end with "STT". This information is usually derived from the SDTM SU domain.

2.3.10 Tumor Load and Metastasis by Location

Tumor load and metastasis are important factors in predicting outcomes for investigational drugs. Knowing the type of metastatic diseases and the tumor burden quantified by Target Lesions Sum of Diameters at baseline and monitoring these over the course of treatment provide highly valuable information regarding drug efficacy.

XX Variable Metadata

Variable Name	Variable Label	Type	Controlled Terms or Format	CDISC Notes
TLSDBL	Baseline Target Lesions Sum of Diam	Num		For example, TR.TRSTRESN where TR.TRTESTCD="SUMLDIAM" and TR.TRGRPID="TARGET" and TR.TRSTAT not equal "NOT DONE" and choose the latest observation where TR.TRDY<=1. Include investigator or independent designation in the definition of the variable to identify who assessed the sum of the diameters for the target lesion(s).
MSBONEFL	Metastasis at Bone Flag	Char	Y, N, null	This usually comes from SDTM domains such as Tumor/Lesion Identification (TU), Findings About Events or Interventions (FA), Medical History (MH), etc.
MSBRNFL	Metastasis at Brain Flag	Char	Y, N, null	This usually comes from SDTM domains such as TU, FA, MH, etc.
MSCHSTFL	Metastasis at Chest Flag	Char	Y, N, null	This usually comes from SDTM domains such as TU, FA, MH, etc.
MSLUNGFL	Metastasis at Lung Flag	Char	Y, N, null	This usually comes from SDTM domains such as TU, FA, MH, etc.
MSLNFL	Metastasis at Lymph Nodes Flag	Char	Y, N, null	This usually comes from SDTM domains such as TU, FA, MH, etc.
MSOTHRFL	Metastasis at Other Sites Flag	Char	Y, N, null	This usually comes from SDTM domains such as TU, FA, MH, etc.
MS*	Metastasis at (loc) Flag	Char	Y, N, null	Prefix MS followed by abbreviated location/site and ending with suffix *FL
NMSBL	Baseline Num of Metastatic Organs	Num		Derived by summing all the sites that have metastasis
NMSBLGRy	BL Num of Metastatic Organs - Group y	Char		Grouped as per study specifications

2.3.11 Variables for Interim Analysis and Censoring

Interim analyses are routinely conducted in oncology studies to evaluate the success of the trial and to make key decisions regarding trial continuation or discontinuation. Data from an interim analysis is used for regulatory submissions in case of early trial success. The following are common variables to capture important dates of interest to facilitate interim analysis.

XX Variable Metadata

Variable Name	Variable Label	Type	Controlled Terms or Format	CDISC Notes
LSTALVDT	Date Last Known Alive	Num		Captures the latest known-alive date after randomization from exposure, subject visits, tumor results, disease response, laboratory assessment, vital sign assessments, survival follow-up alive status, and so on.
DTHDT	Date of Death	Num		Date of subject's death. Assigned from DM.DTHDTC
CUTOFFDT	Data Cutoff Date	Num		The cutoff date for the interim analysis
NCTXSDT	Start Date of New Anti-Cancer Therapy	Num		The earliest date converted to numeric from new anti-cancer therapy records. For example, CM.CMSTDTC where CM.CMCAT/CMGRPID represents new anti-cancer therapy (after date imputation if necessary).

2.4 Subject-level Parameters for Analysis Datasets for Baseline Characteristics (ADBASE)

ADBASE (Analysis Datasets for Baseline Characteristics)

PARAMCD	PARAM	AVAL	AVALC	Fragment	CDISC Notes
DSSTYP	Disease Subtype		Value of disease subtype	*STYP	Describes the smaller groups that a type of cancer can be divided into, based on certain characteristics of the cancer cells. Examples would include, but are not limited to HER2 positive or negative, SCLC or NSCLC.
DSSTBL	Disease Status at Baseline		Value of disease status	*STT	Cancers that are not responding to treatment (refractory) or the return of a disease after a period of improvement (relapsed) at baseline
WBCBL	White Blood Cell Count at Baseline	Value of WBC at baseline		*BL	White blood cell count obtained at baseline
RBCBL	Red Blood Cell Count at Baseline	Value of RBC at baseline		*BL	Red blood cell count obtained at baseline
HGBBL	Hemoglobin at Baseline	Value of HGB at baseline		*BL	Hemoglobin levels obtained at baseline
PLTBL	Platelet Count at Baseline	Value of platelet count at baseline		*BL	Platelet count obtained at baseline
DSFyy	Disease Finding yy		Value of disease findings (e.g., leukocytosis, neutropenia, leukopenia, anemia)	DISF	Disease findings, either collected or derived from labs
IMPTYPyy	Immunophenotype of Lymphocytes yy		Value of immunophenotype of lymphocytes	IMPTYP*	A process that uses antibodies to identify cells based on the types of antigens or markers on the surface of the cells. This process is used to help diagnose diseases.
PCTTy	Prior Cell Transplant Type yy		Value of prior cell type	P*	Autologous or allogeneic hematopoietic cell transplantation
NPRECUR	Number of Prior Recurrences	Value of prior recurrences		P	When cancer returns after a period of remission, it is considered a recurrence. Number of recurrences.
NPLTX	Number of Prior Lines of Therapy	Value of prior lines of therapy		P	
TSMS	Time Since Metastatic Disease	Value of time since metastatic disease		TS*	
TSIDX	Time Since Initial Diagnosis	Value of time since initial diagnosis		TS*	
TSPTX	Time Since Prior Therapy	Value of time since prior therapy		TS*	

3 Cycle Derivation (ADCYCLE)

3.1 ADCYCLE Purpose

In oncology clinical trials, patient care is usually organized in a series of cycles representing specific treatment regimens. When patient care is based on a cycle calendar (e.g., every 21 to 28 days, other interval), a cycle dataset such as ADCYCLE may be used to derive the dates used for each cycle and a common description for the cycle. The dates derived in the ADCYCLE dataset can be used to assign the cycle number and text to any planned activity such as lab assessments, treatments, adverse events, or other analyzed data. Approaches to determining the start and stop of a cycle are defined in the study protocol and refined in the statistical analysis plan (SAP).

The use of cycles within a clinical trial allows for events and findings to be harmonized for analysis. Using a derived cycle also enables determining cycle compliance, by ensuring all data needed for assessment occurs within a common cycle. The use of cycles also allows for compliance to protocol assessments including gaps between cycles which exceed the number of lapsed days between cycles.

Protocol-defined activities need to be considered when determining both planned and actual cycle numbers. Because of this, an interim dataset may be desired to maintain consistency across the other analysis datasets and show how the cycles are being determined. For planned cycles, the dates for coverage would normally be included in the protocol with definitions for how missed cycles should be reported. The actual cycle calculation would be defined in the SAP with details on what defines the start and end of a cycle. By using a interim dataset for cycle definition, other analysis datasets can be created and aligned to a cycle definition by comparing to ADCYCLE. This provides traceability to how the cycles were calculated and helps explain why a derived cycle may be different than what was reported for the subject.

3.2 ADCYCLE Metadata

This model for ADCYCLE follows the principals of the BDS. Each variable for the BDS is defined for the purpose of providing an example. In addition, a separate section is provided to help in setting a series of parameters that can be used in ADCYCLE.

ADCYCLE Variable Metadata

Variable Name	Variable Label	Type	Codelist/Controlled Terms	Core	CDISC Notes
STUDYID	Study Identifier	Char		Req	SDTM.DM.STUDYID
USUBJID	Unique Subject Identifier	Char		Req	SDTM.DM.USUBJID
ASTDT	Analysis Start Date	Num		Perm	This is the start date that each cycle begins, as defined in the protocol. For the purpose of cycle determination this is required.
AENDT	Analysis End Date	Num		Perm	This is the end date that completes each cycle, as defined in the protocol. For the purpose of cycle determination this is required. This should not be blank and Analysis Start Date of the next sequential record used as there may be gaps in the calendar dates that would be missed.
PARAM	Parameter	Char	Actual Cycle, Planned Cycle		The ADCYCLE dataset should have at least 1 record for each subject expected to receive treatment in an oncology trial. At least 1 record should note when each cycle starts for a given subject as defined in the protocol. In addition, the study may also create parameters for recording maximum number of cycles, average cycle duration, or number of missed cycles.
PARAMCD	Parameter Code	Char	ACTCYC, PLNCYC	Req	PARAMCD as described in the ADaMIG.
AVALC	Analysis Value (Char)	Char		Cond	This is the actual number for the cycle reference in the parameter, represented as a text value.
AVAL	Analysis Value	Num		Cond	This is the actual number for the cycle reference in the parameter.

3.3 ADCYCLE Parameters

The following are suggested parameters that can be used in ADCYCLE derivation. Each parameter includes a recommended PARAMCD as well as the intended purpose for each parameter. This is not an all-inclusive list; individual studies may require unique parameters needed for cycle assessments. This list provides a starting point that a programmer or statistician should review in determining which parameters to include.

PARAM	PARAMCD	Definition
Actual Cycle	ACTCYC	The cycle, as derived based on the activity, as defined in the protocol, each subject has performed during the study. There is 1 record for each subject and cycle for when treatment occurred per protocol.
Collected Cycle	COLCYC	The cycle, as collected by the CRF during the conduct of the study, for each subject. This is 1 record for each subject and cycle as reported by the site or the subject.
Planned Cycle	PLNCYC	The cycle as calculated based solely on the protocol and expected activities by the subject during the study. There is 1 record for each subject and cycle combination.
Scheduled Cycles	SCHDCYC	The total number of cycles that a subject should have completed during a study as per protocol. There is 1 record for each subject.
Completed Cycles	COMPCYC	The total number of cycles that a subject actually completed during a study. There is 1 record for each subject.
Missed Cycles	MISSCYC	This is the total number of cycles that a subject did not complete during a study. Often this is the difference between planned cycles and actual cycles. There is 1 record for each subject.
Average Cycle Days	DAYSCYC	This is the average number of days within each cycle that a subject completed, calculated for each cycle as AENDT - ASTDT+1. There is 1 record for each subject

3.4 ADCYCLE Example

When deriving ADCYCLE, the first place to begin is with the Exposure (EX) domain. In this example, subject 100-1001 received drug A on 6 occasions. The VISIT column indicates that at least 1 planned visit was probably missed. The dataset also shows that all dosing was completed on the day of dosing.

ex.xpt

Row	STUDYID	DOMAIN	USUBJID	EXSEQ	EXTRT	EXDOSE	EXDOSU	EXDOSFRM	EXDOSFRQ	EXROUTE	VISIT	EXSTDTC	EXENDTC	EXSTDY	EXENDY
1	ABC100	EX	100-1001	1	DRUG A	1000	mg	INJECTION	ONCE	SUBCUTANEOUS	C1D1	2013-01-03	2013-01-03	1	1
2	ABC100	EX	100-1001	2	DRUG A	1000	mg	INJECTION	ONCE	SUBCUTANEOUS	C2D1	2013-01-24	2013-01-24	22	22
3	ABC100	EX	100-1001	3	DRUG A	1000	mg	INJECTION	ONCE	SUBCUTANEOUS	C3D1	2013-02-14	2013-02-14	43	43
4	ABC100	EX	100-1001	4	DRUG A	1000	mg	INJECTION	ONCE	SUBCUTANEOUS	C4D1	2013-03-06	2013-03-06	63	63
5	ABC100	EX	100-1001	5	DRUG A	1000	mg	INJECTION	ONCE	SUBCUTANEOUS	C6D1	2013-04-15	2013-04-15	103	103
6	ABC100	EX	100-1001	6	DRUG A	1000	mg	INJECTION	ONCE	SUBCUTANEOUS	C7D1	2013-05-06	2013-05-06	124	124

What the preceding example does *not* show is the duration between each dosing or when the actual visits were planned. It also does not readily show when a 21-day cycle is followed, such as in visit 3, or when a visit is more than 21 days apart, such as in visits 4 and 5. It is also not easy to identify when the actual treatment cycle occurs off schedule (which may occur due to unplanned events).

The following ADCYCLE example shows that the subject completed 6 of 7 planned treatments. In addition, the treatments for cycles 4, 6, and 7 occurred prior to the planned treatment date. Showing both planned and actual cycles provides support for the number of missed cycles and the average duration of cycles. This in turn enables adding analysis flags, ANLzzFL, to show when a planned cycle has been skipped or when the duration of a cycle exceeds the maximum allowed days for a cycle.

adcycle.xpt

Row	STUDYID	USUBJID	TRTP	ASTDT	AENDT	PARAMETER	PARAMCD	AVALC	AVAL	EXSEQ
1	ABC100	100-1001	DRUG A	2013-01-03	2013-01-23	Planned Cycle	PLNCYC	Cycle 1	1	
2	ABC100	100-1001	DRUG A	2013-01-24	2013-02-13	Planned Cycle	PLNCYC	Cycle 2	2	
3	ABC100	100-1001	DRUG A	2013-02-14	2013-03-06	Planned Cycle	PLNCYC	Cycle 3	3	
4	ABC100	100-1001	DRUG A	2013-03-07	2013-03-27	Planned Cycle	PLNCYC	Cycle 4	4	
5	ABC100	100-1001	DRUG A	2013-03-28	2013-04-17	Planned Cycle	PLNCYC	Cycle 5	5	
6	ABC100	100-1001	DRUG A	2013-04-18	2013-05-08	Planned Cycle	PLNCYC	Cycle 6	6	
7	ABC100	100-1001	DRUG A	2013-05-09	2013-05-29	Planned Cycle	PLNCYC	Cycle 7	7	
8	ABC100	100-1001	DRUG A	2013-01-03	2013-01-23	Actual Cycle	ACTCYC	Cycle 1	1	1
9	ABC100	100-1001	DRUG A	2013-01-24	2013-02-13	Actual Cycle	ACTCYC	Cycle 2	2	2
10	ABC100	100-1001	DRUG A	2013-02-14	2013-03-05	Actual Cycle	ACTCYC	Cycle 3	3	3
11	ABC100	100-1001	DRUG A	2013-03-06	2013-03-26	Actual Cycle	ACTCYC	Cycle 4	4	4
12	ABC100	100-1001	DRUG A	2013-04-15	2013-05-05	Actual Cycle	ACTCYC	Cycle 6	6	5
13	ABC100	100-1001	DRUG A	2013-05-06	2013-05-06	Actual Cycle	ACTCYC	Cycle 7	7	6
14	ABC100	100-1001	DRUG A			Scheduled Cycles	SCHDCYC	7	7	
15	ABC100	100-1001	DRUG A			Completed Cycles	COMPCYC	6	6	
16	ABC100	100-1001	DRUG A			Missed Cycles	MISSCYC	1	1	
17	ABC100	100-1001	DRUG A			Average Cycle Days	DAYSCYC	17.5	17.5	

By joining the ADCYCLE table with the dataset being analyzed, it is easier to assign actual cycles to other ADaM datasets for additional analysis (e.g., the use of a specified concomitant medication expected for each treatment). Note that for Actual Cycle 7, the end date is the same as the last date since no further visit information is available in this example. If there were a Disposition record with withdrawal information, then end date could be derived using that. Without further data, you would not want to extend beyond that known date of dose.

4 Laboratory Results (ADLB)

4.1 ADLB Purpose

ADaMIG v1.2 introduced bi-direction toxicity variables (see Section 4.9, Table 4.9.1, for an example of how these variables can be utilized). The purpose of this section is to demonstrate the use of such variables in lab reporting with more complex examples.

4.2 Bi-directional Toxicity Grading

4.2.1 ADLB Background

In the ADaMIG, the examples associated with SHIFTy are based on the following pairs (BASECATy, AVALCATy), (BNRIND, ANRIND), (ByIND, AyIND), (BTOXGR, ATOXGR), (BTOXGRL, ATOXGRL), (BTOXGRH, ATOXGRH), (BASE, AVAL), or (BASEC, AVALC). In the example illustrated here, 1 of the predefined paired variables used for defining SHIFTy is not adequate. New variables are derived in order to create paired variables so that the following shift table could be produced.

Table 1 shows an example of the report of the shift in laboratory toxicity grade from baseline to the worst post-baseline value. This section illustrates how to generate this report using variables from the ADaMIG.

Table 1: Example of a Shift in Lab Toxicity Table

Laboratory Toxicity Shift from Baseline to Worst CTCAE Grade Post-baseline Safety Population								
Lab Parameter	Treatment A							
	(N = xxx)							
	Baseline CTCAE Grade	Baseline	Worst Post-baseline CTCAE Grade					
		n (%)	Total n (%)	Grade 0	Grade 1	Grade 2	Grade 3	Grade 4
Parameter		xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)
(units)	Grade 0	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)
<Low/High> ^a	Grade 1	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)
	Grade 2	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)
	Grade 3	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)
	Grade 4	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)
...								
N = number of subjects in safety population; n = number of subjects with non-missing data. <i>High</i> refers to any value above the normal range (i.e., on the hyper side) when table is using BTOXGRL and ATOXGRL. <i>Low</i> refers to any value below the normal range (i.e., on the hypo side) when table is using BTOXGRH and ATOXGRH.								
^a Values at the <Low/High> end of possible range for the parameter are graded with CTCAE Version 4.03 criteria.								

Note: This is not a typical layout for a shift table. This example distinguishes between grade 0 (i.e., normal) and out of range, in the opposite direction of which the toxicity is assessed. How should the current bi-directional toxicity variables be used in this situation in order to remain compliant?

Table 2 illustrates annotation of the table shell using the bi-directional toxicity variables. However, the results in the rows and columns highlighted in blue cannot be easily generated using existing variables.

Table 2: Table 1 with Annotations

Laboratory Toxicity Shift from Baseline to Worst CTCAE Grade Post-baseline Value Safety Population								
Lab Parameter	Treatment A							
	(N = xxx)							
	Baseline CTCAE	Baseline	Worst Post-Baseline CTCAE Grade ATOXGRL / ATOXGRH					
	Grade	n (%)	<High/Low> ^a	Grade 0	Grade 1	Grade 2	Grade 3	Grade 4
ATOXDSCL / ATOXDSCH	BTOXGRL / BTOXGRH							
Parameter	<High/Low>	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)
(units)	Grade 0	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)
<Low/High> ^b	Grade 1	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)
	Grade 2	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)
	Grade 3	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)
	Grade 4	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)
...								

N = number of subjects in safety population; n = number of subjects with non-missing data. Use BTOXGRL and ATOXGRL when table is based on CTC grading in the below-normal range direction. Use BTOXGRH and ATOXGRH when table is based on CTC grading in the above-normal range direction. *High* refers to any value above the normal range (i.e., on the hyper side) when table is using BTOXGRL and ATOXGRL. *Low* refers to any value below the normal range (i.e., on the hypo side) when table is using BTOXGRH and ATOXGRH.

^aThe number of subjects for whom the values of the given lab test are abnormal and are at the opposite end of the range to that defined by the particular CTCAE criteria being applied. ^bValues at the <Low/High> end of possible range for the parameter are graded with CTCAE Version 4.03 criteria.

The lab tests for aspartate aminotransferase (AST), platelet (PLAT), and hemoglobin (HGB) are used to demonstrate how to implement the bi-direction toxicity variables, with the focus being on HGB because of the bi-directional grading. The grade variables are calculated using CTCAE (available at <https://www.eortc.be/services/doc/ctc/>). For the lab test HGB assessed in the low direction, the toxicity is calculated using CTCAE anemia grading (see Table 3). For HGB assessed in the high direction (Table 4), the toxicity is calculated using the CTCAE grading for hemoglobin increased.

Table 3: CTCAE Grading for Anemia – Hemoglobin Assessed in the Low-range Direction

CTCAE v4.03 Term	Grade 1	Grade 2	Grade 3	Grade 4	Grade 5
Anemia	Hemoglobin (Hgb) <LLN - 10.0 g/dL; <LLN - 6.2 mmol/L; <LLN - 100 g/L	Hgb <10.0 - 8.0 g/dL; <6.2 - 4.9 mmol/L; <100 - 80g/L	Hgb <8.0 g/dL; <4.9 mmol/L; <80 g/L; transfusion indicated	Life-threatening consequences; urgent intervention indicated	Death
Definition: A disorder characterized by a reduction in the amount of hemoglobin in 100 ml of blood. Signs and symptoms of anemia may include pallor of the skin and mucous membranes, shortness of breath, palpitations of the heart, soft systolic murmurs, lethargy, and fatigability.					

Table 4: CTCAE Grading for Hemoglobin Increased – Hemoglobin Assessed in the High-range Direction

CTCAE v4.03 Term	Grade 1	Grade 2	Grade 3	Grade 4	Grade 5
Hemoglobin increased	Increase in >0 - 2 gm/dL above ULN or above baseline if baseline is above ULN	Increase in >2 - 4 gm/dL above ULN or above baseline if baseline is above ULN	Increase in >4 gm/dL above ULN or above baseline if baseline is above ULN	-	-
Definition: A finding based on laboratory test results that indicate increased levels of hemoglobin in a biological specimen.					

Grades for other lab tests are calculated similarly, based on the corresponding definition in the CTCAE.

4.2.2 Data

With the use of the bi-directional toxicity variables, both the low and high toxicity grading for lab tests that are assessed in both directions can be captured on the same record, as shown in Table 5.

Table 5: Lab Data Using Bi-directional Toxicity Variables

Row	USUBJID	PARAMCD	VISITNUM	AVAL	ANRLO	ANRHI	ANRIND	BASE	ABLFL	ATOXGRL	ATOXGRH	BNRIND	BTOXGRL	BTOXGRH
1	001-0001	HGB	1	7.4	11	16.1	LOW	7.4	Y	Grade 3	Grade 0	LOW	Grade 3	Grade 0
2	001-0001	HGB	2	20.5	11	16.1	HIGH	7.4		Grade 0	Grade 3	LOW	Grade 3	Grade 0
3	001-0001	AST	1	33	5	25	HIGH	33	Y		Grade 1	HIGH		Grade 1
4	001-0001	AST	2	55	5	25	HIGH	33			Grade 2	HIGH		Grade 1
5	001-0001	PLAT	1	250	150	450	NORMAL	250	Y	Grade 0		NORMAL	Grade 0	
6	001-0001	PLAT	2	100	150	450	LOW	250		Grade 1		NORMAL	Grade 0	
7	001-0002	HGB	1	21.1	11	16.1	HIGH	21.1	Y	Grade 0	Grade 3	HIGH	Grade 0	Grade 3

Because hemoglobin is assessed in both low and high directions, the variables in both directions will be populated. For platelets (PARAMCD = PLAT), because there is CTC grading only in the low direction, only BTOXGRL, ATOXGRL, and other toxicity variables in the low direction are populated; no variables in the high direction can be populated for PARAMCD = PLAT. Even when the value for this parameter is out of range in the high direction none of the HIGH direction toxicity variables (e.g., ANRIND = HIGH for PARAMCD = PLAT) will be populated.

4.2.3 Determining Shift from Baseline

The shift variables can be used to derive a shift in toxicity grade. As illustrated in Table 6, ATOXDSCL is the description of the toxicity in the low direction being assessed; it should be populated if AVAL is not null and if grading is in the low direction, regardless of whether ATOXGRL and BTOXGRL are populated. Similarly, ATOXDSCH is the description of the toxicity in the high direction being assessed and it should be populated if AVAL is not null and if grading is in the high direction, regardless of whether ATOXGRH and BTOXGRH are populated.

Table 6: Example Use of Shift Variables

Row	USUBJID	PARAMCD	VISITNUM	AVAL	ANRLO	ANRHI	ANRIND	BASE	ABLFL	ATOXGRL	ATOXGRH	BNRIND	BTOXGRL	BTOXGRH	ATOXDSCL	ATOXDSCH	SHIFT1	SHIFT2
1	001-0001	HGB	1	7.4	11	16.1	LOW	7.4	Y	Grade 3	Grade 0	LOW	Grade 3	Grade 0	Anemia	Hemoglobin increased		
2	001-0001	HGB	2	20.5	11	16.1	HIGH	7.4		Grade 0	Grade 3	LOW	Grade 3	Grade 0	Anemia	Hemoglobin increased	Grade 3 to Grade 0	Grade 0 to Grade 3
3	001-0001	AST	1	33	5	25	HIGH	33	Y		Grade 1	HIGH		Grade 1		Aspartate aminotransferase increased		
4	001-0001	AST	2	55	5	25	HIGH	33			Grade 2	HIGH		Grade 1		Aspartate aminotransferase increased		Grade 1 to Grade 2
5	001-0001	PLAT	1	250	150	450	NORMAL	250	Y	Grade 0		NORMAL	Grade 0		Platelet count decreased			
6	001-0001	PLAT	2	100	150	450	LOW	250		Grade 1		NORMAL	Grade 0		Platelet count decreased		Grade 0 to Grade 1	
7	001-0002	HGB	1	21.1	11	16.1	HIGH	21.1	Y	Grade 0	Grade 3	HIGH	Grade 0	Grade 3	Anemia	Hemoglobin increased		

Note. Row numbers correspond to those in Table 5.

In this example, SHIFT1 represents the shift from BTOXGRL to ATOXGRL, whereas SHIFT2 represents the shift from BTOXGRH to ATOXGRH. However, in order to produce the desired output displayed in [Table 1](#) for parameters that have bi-directional toxicity grading, determining the shift from baseline in both directions needs to not only look at the corresponding -TOXGRL/-TOXGRH, but also needs to consider ANRIND/BNRIND.

Based on the sample data, there are 2 approaches that can be used to populate SHIFTy:

1. Create variables that combine the values of -TOXGRL and -NRIND and -TOXGRH and -NRIND so that toxicity grading and out of range in the opposite direction is contained in a single variable instead of 2 separate variables.
2. Incorporate the logic into a single programming statement.

4.2.3.1 Creating New Variables

Table 7 illustrates the first approach, creating a set of variables based on the baseline and analysis variables.

Table 7: Variables Used for Populating SHIFTy

Variable	Description
BTOXDRL	If ATOXDSCl is populated, then create BTOXDRL and set to BTOXGRL if BTOXGRL contains a non-missing value; otherwise set to BNRIND.
ATOXDRL	If ATOXDSCl is populated, then create ATOXDRL is set to ATOXGRL if ATOXGRL contains a non-missing value; otherwise set to ANRIND.
BTOXDIRH	If ATOXDSCl is populated, then create BTOXDIRH and set to BTOXGRH if BTOXGRH contains a non-missing value; otherwise set to BNRIND.
ATOXDIRH	If ATOXDSCl is populated, then create ATOXDIRH and set to ATOXGRH if ATOXGRH contains a non-missing value; otherwise set to ANRIND.

Shift in toxicity in the low direction:

Per update to ADaMIG, the shift variables can be based on any baseline and current record variable pairs. Therefore, SHIFT1 is set to a concatenation of BTOXDRL, the word "to" and ATOXDRL.

Shift in toxicity in the high direction:

SHIFT2 is set to a concatenation of BTOXDIRH, the word "to" and ATOXDIRH.

Table 8: Example Use of Proposed -TOXDRL and -TOXDIRH Variables

Row	USUBJID	PARAMCD	VISITNUM	BTOXDRL	ATOXDRL	BTOXDIRH	ATOXDIRH	SHIFT1	SHIFT2
1	001-0001	HGB	1	Grade 3	Grade 3	Low	Low	Grade 3 to Grade 3	Low to Low
2	001-0001	HGB	2	Grade 3	High	Low	Grade 3	Grade 3 to High	Low to Grade 3
3	001-0001	AST	1		High	Grade 1	Grade 1		Grade 1 to Grade 1
4	001-0001	AST	2		High	Grade 1	Grade 2		Grade 1 to Grade 2
5	001-0001	PLAT	1	Grade 0	Grade 0			Grade 0 to Grade 0	
6	001-0001	PLAT	2	Grade 0	Grade 1			Grade 0 to Grade 1	
7	001-0002	HGB	1	High	High	Grade 3	Grade 3	High to High	Grade 3 to Grade 3

4.2.3.2 Using Programming Logic

An alternative to creating 4 new variables in the dataset is to incorporate the logic into a single programming statement. The COALESCEC function assigns the value of the first non-missing character variable value in the order in which the variables are listed. Each coalescec function represents a temporary variable. The CATX function concatenates the 2 values returned by the COALESCEC functions separated by the specified delimiter (i.e., "to").

SHIFT1 = catx("to", coalescec(BTOXGRL, propcase(BNRIND)), coalescec(ATOXGRL, propcase(ANRIND)));

SHIFT2 = catx("to", coalescec(BTOXGRH, propcase(BNRIND)), coalescec(ATOXGRH, propcase(ANRIND)));

Although the programming logic is illustrated using SAS, similar logic could be implemented with other software.

4.2.4 Disclaimer

Concepts and variables introduced with either of these approaches for determining the shift for the bi-directional toxicity grading are not in the ADaMIG and are being proposed. Therefore, any new variable or concept should be used with caution and with the understanding that it may not be adopted or may be slightly modified.

4.2.5 Rationale

- ATOXDSCH and ATOXDSCL allow determining eligibility for assessment for CTC grading. In the preceding example, at baseline, subject 001 had AVAL with "Anemia" grade 3 and subject 002 had AVAL with "Hemoglobin increased" grade 3. Both subjects are eligible for assessment on the anemia table and on the Hemoglobin increased table.
- SHIFT1 and SHIFT2 are populated based on a combination of BTOXGR-, ATOXGR- and ANRIND, any of which can be null. In the example in [Table 8](#), row 2 has BTOXGRL populated but ATOXGRL is null. One would still populate SHIFT1 as "Grade 3 to High" to show that it went from grade 3 to high.

5 Study Drug Exposure (ADEXP and ADEXSUM)

5.1 ADEXP and ADEXSUM Purpose

This section is intended to assist in creating ADaM datasets for the exposure-related domain(s) in oncology studies and is to be used as a reference document in conjunction with ADaMIG v1.2. The level of detail of exposure analyses varies depending on therapeutic area needs, clinical phase, and others. This guideline acknowledges the need for flexibility and therefore recommendations made in the following sections are to be seen as guidance only.

5.1.1 Sample Exposure Table

The following table provides an example to facilitate understanding of exposure analysis. The Exposure table shows the types of analysis expected when submitting to a regulatory agency for an oncology study. The exposure calculations on this page are also typical of a compound administered through infusion to the subjects. For any study, the actual Exposure table should be created based on the analysis expected as described in the analysis plan.

	Treatment 1 (N = XXX)		Treatment 2 (N = XXX)	Treatment 3 (N = XX)
	Component A	Component B		
Treatment Duration (months)				
n	xxx	xx	xxx	xx
Mean (SD)	x.x (x.x)	x (x)	x.x (x.x)	x.x (x.x)
Median	x.x	x	x.x	x.x
Range	x.xx to xx	x.xx to xx.x	x.xx to xx.x	x.xx to xx.x
n (%)	xxx	xx	xxx	xx
<3 Months	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)
>=3 Months	xx (xx.x)	xx (xx.x)	xx (xx.x)	xx (xx.x)
Number of Administrations				
n	xxx	xx	xxx	xx
Mean (SD)	xx (xxx.x)	x (x.x)	x.x (x.x)	xx (xx.x)
Median	xx.x	x	x	xx.x
Range	x to xxx	x to xx	x to xx	x to xxx
Number of Reductions				
N	xxx	xx	xxx	xx
Mean (SD)	xx (xxx.x)	x (x.x)	x.x (x.x)	xx (xx.x)
Median	xx.x	x	x	xx.x
Range	x to xxx	x to xx	x to xx	x to xxx
Relative Dose Intensity				
n	xxx	xx	xxx	xx
Mean (SD)	xx (xxx.x)	x (x.x)	x.x (x.x)	xx (xx.x)
Median	xx.x	x	x	xx.x
Range	x to xxx	x to xx	x to xx	x to xxx
Number of Delays				
n	xxx	xx	xxx	xx
Mean (SD)	xx (xxx.x)	x (x.x)	x.x (x.x)	xx (xx.x)
Median	xx.x	x	x	xx.x
Range	x to xxx	x to xx	x to xx	x to xxx
Number of Interruptions				
n	xxx	xx	xxx	xx
Mean (SD)	xx (xxx.x)	x (x.x)	x.x (x.x)	xx (xx.x)
Median	xx.x	x	x	xx.x
Range	x to xxx	x to xx	x to xx	x to xxx

5.2 Exposure Analysis Dataset (ADEXP)

Structure: One record per subject per treatment (EXTRT) per start date

In oncology studies, it is common to create an intermediate dataset using SDTM domains (e.g., EX/SUPPEX, EC/SUPPEC), adding variables from ADSL, and deriving new variables. An intermediate dataset may not be used for generating analytical outputs but could be used for listing purposes. The dataset (ADEXP in this example) follows general ADaM principles (e.g., naming conventions for derived variables) and provides metadata by describing derivations and data-point traceability by carrying forward EXSEQ from the Exposure (EX) domain as the primary location but documenting where EC data may be used (e.g., for flags). The name of the ADEXP dataset was intentionally chosen to avoid the perception that EX should be its only source.

- Note:
- Although ADEXP does not use the BDS structure, it still follows the fundamental ADaM principles.
 - This example, for illustration purposes, uses the SDTM EX domain. Other SDTM domains that fulfill the expected need could be used (e.g., Exposure as Collected (EC)).

Relevant study-drug exposure data can be found in various SDTM domains and are a direct representation of what was documented by investigators on the CRF. Study design, dosing scheme, number of drugs and nature of treatments, route of administration, and other factors determine how data are captured on the CRF and represented in SDTM domains. The derivation of analysis parameters directly from SDTM domains can be challenging when exposure data and their representation in SDTM are complex. The optional dataset ADEXP can be helpful to pool data from different SDTM domains in an intermediate analysis dataset for the derivation of analysis parameters. The decision to use ADEXP as well as the ADaM class of the dataset is at the sponsor's discretion.

One record in ADEXP represents the administration of (study) drug at an unchanged dose level during the time window defined by ASTDT and AENDT where AVAL is the actual total dose taken during the time window. The following is an example of an ADEXP structure that uses 1 record per time window. EX and EC domains are likely inputs to ADEXP.

In this example, subjects are supposed to have a weekly infusion of drug Z. As shown in row 3, subject ABC123-0201 was planned to have less dosage because of dose-limiting toxicity (DLT; see ECADJ). Row 6 shows a missed cycle 3 visit. Row 8 indicates an interrupted infusion because of an injection site reaction (EXADJ). Per the study protocol, a gap of more than 9 days between infusions is considered a dose delay. The cycle 5 infusion was delayed, occurring on day 32 rather than day 29. A delay of more than 2 days is considered a delay.

ec.xpt

Row	STUDYID	DOMAIN	USUBJID	ECSEQ	ECLNKID	ECLNKGRP	EXTRT	ECMOOD	ECPRESP	ECOCUR	ECDOSE	ECDOSU	ECPSTRG	ECPSTRGU	ECADJ	VISITNUM	VISIT	EPOCH	ECSTDTC	ECENDTC	ECSTDY	ECENDY
1	ABC123	EC	ABC123-0201	1		V1	DRUG Z	SCHEDULED			10	mg/kg				1	Cycle 1 Day 1	TREATMENT	2009-02-13	2009-02-13	1	1
2	ABC123	EC	ABC123-0201	2	20090213T1000	V1	DRUG Z	PERFORMED	Y	Y	100	mL	5.5	mg/mL		1	Cycle 1 Day 1	TREATMENT	2009-02-13T10:00	2009-02-13T10:45	1	1
3	ABC123	EC	ABC123-0201	3		V2	DRUG Z	SCHEDULED			7.5	mg/kg			Dose limiting toxicity	2	Cycle 2 Day 1	TREATMENT	2009-02-20	2009-02-20	8	8
4	ABC123	EC	ABC123-0201	4	20090220T1100	V2	DRUG Z	PERFORMED	Y	Y	75	mL	5.5	mg/mL		2	Cycle 2 Day 1	TREATMENT	2009-02-20T11:00	2009-02-20T11:40	8	8
5	ABC123	EC	ABC123-0201	5		V3	DRUG Z	SCHEDULED			7.5	mg/kg				3	Cycle 3 Day 1	TREATMENT	2009-02-27	2009-02-27	15	15
6	ABC123	EC	ABC123-0201	6	20090227	V3	DRUG Z	PERFORMED	Y	N		mL	5.5	mg/mL		3	Cycle 3 Day 1	TREATMENT	2009-02-27	2009-02-27	15	15
7	ABC123	EC	ABC123-0201	7		V4	DRUG Z	SCHEDULED			7.5	mg/kg				3	Cycle 4 Day 1	TREATMENT	2009-03-06	2009-03-06	22	22
8	ABC123	EC	ABC123-0201	8	2009020306	V4	DRUG Z	PERFORMED	Y	Y	50	mL	5.5	mg/mL		3	Cycle 4 Day 1	TREATMENT	2009-03-06T11:00	2009-03-06T11:20	22	22
9	ABC123	EC	ABC123-0201	9		V5	DRUG Z	SCHEDULED			7.5	mg/kg				3	Cycle 5 Day 1	TREATMENT	2009-03-16	2009-03-16	32	32

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Row	STUDYID	DOMAIN	USUBJID	ECSEQ	ECLNKID	ECLNKGRP	ECTRT	ECMOOD	ECPRESP	EOCCUR	ECDOSE	ECDOSU	ECPSTRG	ECPSTRGU	ECADJ	VISITNUM	VISIT	EPOCH	ECSTDTC	ECENDTC	ECSTDY	ECENDY
10	ABC123	EC	ABC123-0201	10	2009020301	V5	DRUG Z	PERFORMED	Y	Y	75	mL	5.5	mg/mL		3	Cycle 5 Day 1	TREATMENT	2009-03-16T11:00	2009-03-16T11:42	32	32

Row	STUDYID	RDOMAIN	USUBJID	IDVAR	IDVARVAL	QNAM	QLABEL	QVAL	QORIG	QEVAL
1	ABC	EC	ABC123-0201	ECSEQ	6	ECREASOC	Reason for Occur Value	PERSONAL REASON	CRF	

ex.xpt

Row	STUDYID	DOMAIN	USUBJID	EXSEQ	EXLNKID	EXLNKGRP	EXTRT	EXDOSE	EXDOSU	EXDOSFRM	EXDOSFRQ	EXROUTE	EXADJ	VISITNUM	VISIT	EPOCH	EXSTDTC	EXENDTC	EXSTDY	EXENDY
1	ABC123	EX	ABC123-0201	1	20090213T1000	V1	DRUG Z	10	mg/kg	SOLUTION	CONTINUOUS	INTRAVENOUS		1	Cycle 1 Day 1	TREATMENT	2009-02-13T10:00	2009-02-13T10:00	1	1
2	ABC123	EX	ABC123-0201	2	20090220T1100	V2	DRUG Z	7.5	mg/kg	SOLUTION	CONTINUOUS	INTRAVENOUS		2	Cycle 2 Day 1	TREATMENT	2009-02-20T11:00	2009-02-20T11:00	8	8
3	ABC123	EX	ABC123-0201	2	2009020306	V4	DRUG Z	5	mg/kg	SOLUTION	CONTINUOUS	INTRAVENOUS	Injection site reaction	4	Cycle 4 Day 1	TREATMENT	2009-03-06	2009-03-06	22	22
4	ABC123	EX	ABC123-0201	2	2009020316	V5	DRUG Z	7.5	mg/kg	SOLUTION	CONTINUOUS	INTRAVENOUS		5	Cycle 5 Day 1	TREATMENT	2009-03-16	2009-03-16	32	32

ADEXP Variable Metadata

Variable Name	Variable label	Type	Codelist/Controlled Terms	Core	CDISC Notes
EXTRT	Name of Treatment	Char		Req	EX.EXTRT
EXDOSE	Dose	Num		Req	EX.EXDOSE
EXDOSTXT	Dose Description	Char		Perm	EX.EXDOSTXT
EXDOSEU	Dose Unit	Char		Req	EX.EXDOSEU
ECDOSP	Planned Dose	Num		Cond	Planned Dose could be obtained from protocol or SAP for that visit or it might be available in EC domain.
ECDOSPU	Planned Dose Unit			Cond	Unit corresponding to ECDOSP.
EXDOSFRM	Dose Form	Char		Cond	EX.EXDOSEFRM
EXDOSFRQ	Dosing Frequency Per Interval	Char		Cond	EX.EXDOSEFRQ
EXDOSRGM	Intended Dose Regimen	Char		Cond	EX.EXDOSRGM
EXROUTE	Route of Administration	Char		Req	EX.EXROUTE
EXADJ	Reason for Dose Adjustment	Char		Req	EX.EXADJ
EXSTDTC	Start Date/Time of Treatment	Char		Req	EX.EXSTDTC
EXENDTC	End Date/Time of Treatment	Char		Req	EX.EXENDTC
EXSTDY	Study Day of Start of Treatment	Num		Req	EX.EXSTDY
EXENDY	Study Day of End of Treatment	Num		Req	EX.EXENDY
EXSEQ	Sequence Number	Num		Req	EX.EXSEQ
ASTDT	Analysis Start Date	Num		Req	Created from converting XX.--STDTC from character ISO8601 format to numeric date format.
AENDT	Analysis End Date	Num		Req	Created from converting XX.--ENDTC from character ISO8601 format to numeric date format.
ASTM	Analysis Start Time	Num		Cond	Created from converting XX.--STDTC from character ISO8601 format to numeric time format.
AETM	Analysis End Time	Num		Cond	Created from converting XX.--ENDTC from character ISO8601 format to numeric time format.
EXDUR	Duration of Treatment	Char		Cond	EX.EXDUR
EXDURD	Duration of Treatment (days)	Num		Cond	Duration of Treatment derived from EXSTDTC and EXENDTC in days.
DOSREDFL	Dose Reduced Flag	Char	(NY)	Cond	If EXDOSE < ECDOSEP (convert ECDOSEP into same units used for EXDOSE, if needed)
DOSINTFL	Dose Interrupted Flag	Char	(NY)	Cond	Set to "Y", if EOCCUR="N" or defined in protocol or SAP to have received treatment at a time point but no EX record with treatment
DOSDELFL	Dose Delay Flag	Char	(NY)	Cond	Set to "Y" for all cases if the actual dose falls outside of the allowed dosing window. Use SAP and protocol to define derivation of Dose Delay.
TRTSDT	Date of First Exposure to Treatment	Num		Cond	ADSL.TRITSDT
TRTEDT	Date of Last Exposure to Treatment	Num		Cond	ADSL.TRITEDT

Variable Name	Variable label	Type	Codelist/Controlled Terms	Core	CDISC Notes
TRTSTM	Time of First Exposure to Treatment	Num		Cond	ADSL.TRTSTM
TRTETM	Time of Last Exposure to Treatment	Num		Cond	ADSL.TRTETM
TRTSDTM	Datetime of First Exposure to Treatment	Num		Cond	ADSL.TRTSDTM
TRTEDTM	Datetime of Last Exposure to Treatment	Num		Cond	ADSL.TRTEDTM
TRxxSDT	Date of First Exposure in Period xx	Num		Cond	ADSL.TRxxSDT
TRxxSTM	Time of First Exposure in Period xx	Num		Cond	ADSL.TRxxSTM
TRxxSDTM	Datetime of First Exposure in Period xx	Num		Cond	ADSL.TRxxSDTM
TRxxEDT	Date of Last Exposure in Period xx	Num		Cond	ADSL.TRxxEDT
TRxxETM	Time of Last Exposure in Period xx	Num		Cond	ADSL.TRxxETM
TRxxEDTM	Datetime of Last Exposure in Period xx	Num		Cond	ADSL.TRxxEDTM

In ADEXP row 2 is flagged as DOSREDFL = "Y" because EXDOSE < ECDOSEP. In this case ECDOSEP is based on protocol; it was reduced from 10mg/kg to 7.5 mg/kg due to DLT. Row 3 is flagged as DOSINTFL= "Y" as cycle 3 was missed. This record is obtained from EC domain and does not exist in EX.

Row 5 is flagged as DOSEDELFL = "Y" because it is 10 days after previous dose, which is greater than protocol-defined 9 days. Analysis window variables are used in the following example. In this ADY is greater than AWHI for that visit, hence it is flagged as a dose delay.

adexp.xpt

Row	STUDYID	USUBJID	EXTRT	ASTDT	AENDT	VISIT	EXDOSE	EXDOSU	ECDOSP	ECDOSPU	EXADJ	ECADJ	ECREASOC	ACYCLE	ADY	AWTARGET	AWTDIFF	AWLO	AWHI	AWU	DOSREDFL	DOSINTFL	DOSDELFL
1	ABC123	ABC123-0201	DRUG Z	2009-02-13	2009-02-13	Cycle 1 Day 1	10	mg/kg	10	mg/kg				Cycle 1	1	1	0	1	1	Days			
2	ABC123	ABC123-0201	DRUG Z	2009-02-20	2009-02-20	Cycle 2 Day 1	7.5	mg/kg	10	mg/kg		Dose limiting toxicity		Cycle 2	8	8	0	8	10	Days	Y		
3	ABC123	ABC123-0201	DRUG Z	2009-02-27	2009-02-27	Cycle 3 Day 1			7.5	mg/kg			PERSONAL REASON	Cycle 3	15	15	0	15	17	Days		Y	
4	ABC123	ABC123-0201	DRUG Z	2009-03-06	2009-03-06	Cycle 4 Day 1	5	mg/kg	7.5	mg/kg	Injection site reaction			Cycle 4	22	22	0	22	24	Days	Y		
5	ABC123	ABC123-0201	DRUG Z	2009-03-16	2009-03-16	Cycle 5 Day 1	7.5	mg/kg	7.5	mg/kg				Cycle 5	32	29	3	29	31	Days			Y

5.3 Exposure Summary Analysis Dataset (ADEXSUM)

Structure: One record per subject per parameter per analysis interval

ADEXSUM can be based on various data sources depending on complexity of the trial and sponsor discretion. Intermediate analysis datasets such as ADEXP can feed into ADEXSUM, if available, but it is also possible to derive ADEXSUM completely based on SDTM sources such as EX and EC. ADEXSUM is always in BDS structure and includes the parameters that are frequently required and reported (e.g., Treatment Duration Actual in Days, Treatment Duration Planned in Days, Number of Actual Study Drug Administrations, Number of Actual Cycles, Number of Planned Cycles, Cumulative Planned Dose, Cumulative Actual Dose). The PARQUAL variable is included as parameter qualifier to distinguish analysis values coming from different drugs used in combination therapy, a typical scenario in oncology trials. The timing for ADEXSUM is AEVLINT (Analysis Interval for Evaluation); timing may be based on the entire study period, a subperiod, or cycles.

ADEXSUM Variable Metadata

Variable Name	Variable label	Type	Codelist/Controlled Terms	Core	CDISC Notes
AEVLINT	Analysis Evaluation Interval	Char		Perm	Represents the interval for analysis. This is used to cover an interval of time to calculate the AVAL.
PARQUAL	Parameter Qualifier	Char		Perm	Description of the treatment summarized on each record. Can be equal to EXTRT for summaries/evaluations of individual treatments, or "All" for summaries/evaluations across all treatments.
PARAM	Parameter	Char		Req	The description of the analysis parameter.
PARAMCD	Parameter Code	Char		Req	The short name of the analysis parameter in PARAM.
AVAL	Analysis Value	Num		Cond	Numeric analysis value described by PARAM.
AVALC	Analysis Value (C)	Char		Cond	Character analysis value described by PARAM.
AVALU	Analysis Value Unit	Char		Cond	Unit corresponding to AVAL.

ADEXSUM Value-level Metadata: Parameter Value List - ADEXSUM

AVAL/AVALC	AEVLINT	PARQUAL	PARAMCD	PARAM	CDISC Notes
AVAL	Overall	All	TRTDURD	Treatment Duration Actual in Days	Total duration in days of treatment over all values of EXTRT for each subject
AVAL	Overall	All	TRTPDURD	Treatment Duration Planned in Days	Total planned duration of treatment over all values of EXTRT for each subject.
AVAL	Overall	All	NADMIN	Nr of Actual Study Drug Administrations	Number of treatment administrations for all values of EXTRT combined per subject.
AVAL	Overall	All	NUMCYC	Number of Actual Cycles	Total number of cycles across all treatments per subject. Sponsors should discuss definition of cycles with the review team ahead of submission, if applicable.
AVAL	Overall	All	NUMPCYC	Number of Planned Cycles	Refer to SAP or protocol on how to derive planned cycle.
AVAL	Overall and/or <by cycle>	<each value of EXTRT>	CUMPLDOS	Cumulative Planned Dose	Cumulative planned dose for each treatment for each subject over each evaluation interval (AEVLINT). Unit should be captured in AVALU.
AVAL	Overall and/or <by cycle>	<each value of EXTRT>	CUMACDOS	Cumulative Actual Dose	Cumulative actual dose for each treatment for each subject over each evaluation interval (AEVLINT). Unit should be captured in AVALU.

AVAL/AVALC	AEVLINT	PARQUAL	PARAMCD	PARAM	CDISC Notes
AVAL	Overall and/or <by cycle>	< each value of EXTRT>	RDOSINT	Relative Dose Intensity (%)	Ratio of actual to planned dose expressed as percentage for each subject over each evaluation interval.
AVALC	Overall and/or <by cycle>	<each value of EXTRT>	DELAY	Any Dose Delays	"Y" if there were any dose delays over each evaluation interval.
AVAL	Overall and/or <by cycle>	<each value of EXTRT>	NDELAY	Number of Dose Delays	Total number of dose delays over each evaluation interval
AVAL	Overall and/or <by cycle>	<each value of EXTRT>	DURDLAY	Total Duration of Delays	Total duration of delays. Refer to SAP or protocol on what constitutes a delay and how to delay is derived.
AVALC	Overall and/or <by cycle>	<each value of EXTRT>	DOSRED	Any Dose Reductions	"Y" if there were any dose reductions
AVAL	Overall and/or <by cycle>	<each value of EXTRT>	NDOSRED	Number of Dose Reductions	Number of dose reductions
AVALC	Overall and/or <by cycle>	<each value of EXTRT>	DOSINT	Any Dose Interruptions	"Y" if there were any dose interruptions
AVAL	Overall and/or <by cycle>	<each value of EXTRT>	NDOSINT	Number of Dose Interruptions	Number of dose interruptions
AVALC	Overall	ALL	CYCDLAY	Any Cycle Delays	"Y" if there were any cycle delays
AVAL	Overall	ALL	NCYCDLAY	Number of Cycle Delays	Number of cycle delays
AVALC	Overall	<each value of EXTRT>	DOSDCTS	Any Dose Discontinuations	"Y" if there were any dose discontinuations.
AVAL	Overall	<each value of EXTRT>	NDOSDCTS	Number of Dose Discontinuations	Total number of dose discontinuations
AVALC	Overall	ALL	TRTDCT	Treatment Discontinued	"Y" if treatment permanently discontinued for all drugs

The following table is an ADEXSUM example dataset, derived using the preceding ADEXP example dataset. AVAL/AVALC were computed as follows:

- NADMIN - Counted total number of non-missing ADEXP.EXDOSE records per subject
- NUMCYC - Count unique number of ADEXP.ACYLE where ADEXP.EXDOSE not missing
- NUMPCYC - Count unique number of ADEXP.ACYLE or refer to protocol/SAP on how this parameter should be defined
- CUMPLDOS - For this example added ECDOSEP per subject and EXTRT or refer to protocol/SAP on how this parameter should be defined
- CUMACDOS - Adding up all EXDOSE per subject and EXTRT
- RDOSINT - Ratio of Cumulative Actual Dose to Cumulative Planned Dose in percentage
- NDELAY - Count number of records where ADEXP.DOSDELFL is "Y" for that evaluation interval
- NCYCDLAY - Count number of unique ADEXP.ACYLE records with DOSDELFL is "Y"
- NDOSRED - Count number of records with DOSREDFL is "Y" per subject for the evaluation interval
- NDOSINT - Count number of records with DOSINTFL is "Y" per subject for the evaluation interval

Row	STUDYID	USUBJID	PARAMCD	PARAM	AEVLINT	PARQUAL	AVAL	AVALC	AVALU
1	ABC123	ABC123-0201	TRTDURD	Treatment Duration Actual in Days	Overall	All	32		Days
3	ABC123	ABC123-0201	NADMIN	Nr of Actual Study Drug Administrations	Overall	All	4		
4	ABC123	ABC123-0201	NUMCYC	Number of Actual Cycles	Overall	All	4		
5	ABC123	ABC123-0201	NUMPCYC	Number of Planned Cycles	Overall	All	5		
6	ABC123	ABC123-0201	CUMPLDOS	Cumulative Planned Dose	Overall	Drug Z	42.5		mg/kg
7	ABC123	ABC123-0201	CUMACDOS	Cumulative Actual Dose	Overall	Drug Z	30		mg/kg
8	ABC124	ABC123-0202	RDOSINT	Relative Dose Intensity (%)	Overall	Drug Z	70.6		%
9	ABC125	ABC123-0203	DELAY	Any Dose Delays	Overall	Drug Z		Y	
10	ABC126	ABC123-0204	NDELAY	Number of Dose Delays	Overall	Drug Z	1		
11	ABC127	ABC123-0205	DURDLAY	Total Duration of Delays	Overall	Drug Z	1		days
12	ABC128	ABC123-0206	CYCDLAY	Any Cycle Delays	Overall	All		Y	
13	ABC129	ABC123-0207	NCYCDLAY	Number of Cycle Delays	Overall	All	1		
15	ABC132	ABC123-0210	DOSRED	Any Dose Reductions	Overall	Drug Z		Y	
16	ABC133	ABC123-0211	NDOSRED	Number of Dose Reductions	Overall	Drug Z	2		
17	ABC134	ABC123-0212	DOSINT	Any Dose Interruptions	Overall	Drug Z		Y	
18	ABC135	ABC123-0213	NDOSINT	Number of Dose Interruptions	Overall	Drug Z	1		

Appendices

Appendix A: Glossary and Abbreviations

The following abbreviations and terms are used in this document. Additional definitions can be found in the text, and in the CDISC Glossary available at <https://www.cdisc.org/standards/glossary>.

ADaM	Analysis Dataset Model
ADaMIG	ADaM Implementation Guide
ADBASE	Analysis Dataset for Baseline Data
ADSL	(ADaM) Subject-level Dataset
AJCC	American Joint Committee on Cancer
Antineoplastic therapies	Anti-cancer treatments
BDS	(ADaM) Basic Data Structure
CDISC	Clinical Data Interchange Standards Consortium
Controlled terminology	A set of standard value lists that are used throughout the clinical research process, from data collection through analysis and submission
CRF	Case report form (sometimes "case record form")
CTCAE	Common Terminology Criteria for Adverse Events
CTC	Circulating tumor cell
ECOG	Eastern Cooperative Oncology Group
EVS	(NCI) Enterprise Vocabulary Services
FDA	(US) Food and Drug Administration
Histopathology	Diagnosis and study of diseases of the tissues
IHC	Immunohistochemistry; the process of selectively identifying antigens in cells of a tissue section by exploiting the principle of antibodies binding specifically to antigens in biological tissues
Immunophenotyping	Test used to identify cells on the basis of the types of markers or antigens present on the cell's surface, nucleus, or cytoplasm
IRT	Interactive response technology
KPSS	Karnofsky Performance Status Scale
mCRPC	Metastatic castration-resistant prostate cancer
NCI	National Cancer Institute (NIH)
NIH	(US) National Institutes of Health
OCE	(FDA) Oncology Center of Excellence
OOD	(FDA) Office of Oncologic Diseases
PSA	Prostate-specific antigen
SAP	Statistical analysis plan
SAS	Statistical Analysis System (software)
SDTM	Study Data Tabulation Model
SDTMIG	Study Data Tabulation Model Implementation Guide: Human Clinical Trials
TAUG	Therapeutic area user guide
TNM	Tumor-node-metastasis (staging of cancer)
WHO	World Health Organization

Appendix B: Representations and Warranties, Limitations of Liability, and Disclaimers

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